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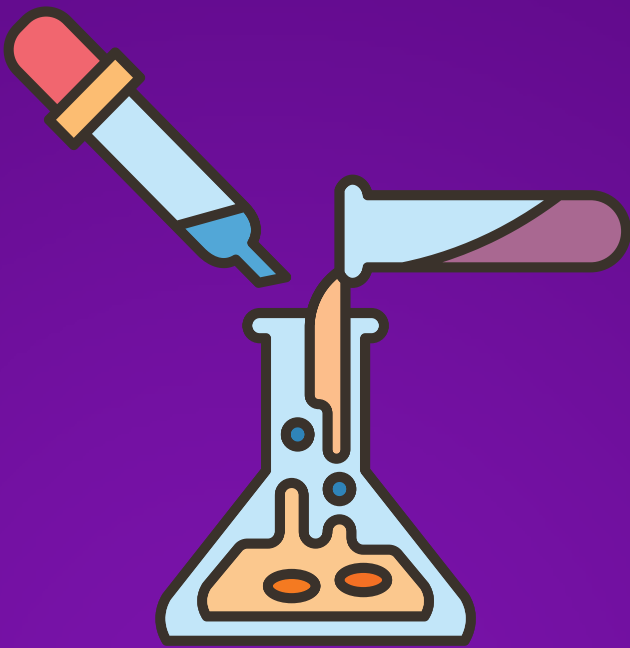

INORGANIC CHEMISTRY

JEE (MAIN+ADVANCED)

ENTHUSIAST COURSE

EXERCISE

Hydrogen and its Compounds

English Medium

EXERCISE # O-1

DIHYDROGEN

1. The sum number of neutrons and protons in one of the isotopes of hydrogen is :-

(A) 3 (B) 4 (C) 5 (D) 6

HY0001

2. The catalyst used in Bosch process of manufacture of H_2 is :-

(A) Finely divided Ni (B) V_2O_5 (C) Pb (D) $Fe_2O_3 + Cr_2O_3$

HY0002

3. The most abundant isotope of hydrogen is :-

(A) Tritium (B) Deuterium (C) Protium (D) Para hydrogen

HY0003

4. The n/p ratio for ${}_1H^1$ is :-

(A) 1 (B) 2 (C) 3 (D) Zero

HY0004

5. Ordinary hydrogen at high temperature is a mixture of :-

(A) 75% o-Hydrogen + 25% p-Hydrogen (B) 25% o-Hydrogen + 75% p-Hydrogen

(C) 50% o-Hydrogen + 50% p-Hydrogen (D) 1% o-Hydrogen + 99% p-Hydrogen

HY0005

6. Hydrogen is behave as :-

(A) Electropositive

(B) Electronegative

(C) Both electropositive as well as elecctro-negative

(D) Neither electropositive nor electronegative

HY0006

7. At high temperature Para hydrogen is :-

(A) Less stable than ortho hydrogen

(B) More stable than ortho hydrogen

(C) As stable as ortho hydrogen

(D) None of these

HY0007

8. When the same amount of zinc is treated separately with excess of sulphuric acid and excess of sodium hydroxide, the ratio of volumes of hydrogen evolved is :-

(A) 1 : 1 (B) 1 : 2 (C) 2 : 1 (D) 9 : 4

HY0008

9. The lightest gas is :-

(A) Nitrogen (B) Helium (C) Oxygen (D) Hydrogen

HY0009

10. The ratio of electron, proton and neutron in tritium is :-

(A) 1 : 1 : 1 (B) 1 : 1 : 2 (C) 2 : 1 : 1 (D) 1 : 2 : 1

HY0010

11. The nuclei of tritium (H^3) atom would contain neutrons :-

(A) 1 (B) 2 (C) 3 (D) 4

HY0011

12. The adsorption of hydrogen by metals is called :-

(A) Dehydrogenation (B) Hydrogenation (C) Occlusion (D) Adsorption

HY0012

13. At absolute zero :-

(A) Only para hydrogen exists (B) Only ortho hydrogen exists

(C) Both para and ortho hydrogen exist (D) None

HY0013

WATER (H_2O)

14. Only temporary hardness in water is removed by :-

(A) Boiling (B) Filtration (C) Calgon's process (D) None of these0

HY0014

15. Both temporary and permanent hardness is removed on boiling water with :-

(A) $Ca(OH)_2$ (B) Na_2CO_3 (C) $CaCO_3$ (D) CaO

HY0015

16. Temporary hardness is caused due to the presence of :-

(A) $CaSO_4$ (B) $CaCl_2$ (C) $CaCO_3$ (D) $Ca(HCO_3)_2$

HY0016

17. High boiling point of water is due to :-

(A) Its high specific heat (B) Hydrogen bonding

(C) High dielectric constant (D) Low dissociation constant

HY0017

18. Calgon is an industrial name given to :-

(A) Normal sodium phosphate (B) Sodium meta-aluminate

(C) Sodium hexametaphosphate (D) Hydrated sodium aluminium silicate

HY0018

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19. Permutit is :-

(A) Hydrated sodium aluminium silicate (B) Sodium hexametaphosphate

(C) Sodium silicate (D) Sodium meta-aluminate

HY0019

20. Heavy water has found application in atomic reactor as :-

(A) Coolant (B) Moderator

(C) Both coolant and moderator (D) Neither coolant nor moderator

HY0020

21. Calgon (a water softener) is :-

(A) $Na_2[Na_4(PO_3)_6]$ (B) $Na_4[Na_2(PO_3)]_6$ (C) $Na_2[Na_4(PO_4)]_6$ (D) $Na_4[Na_2(PO_4)_6]$

HY0021

22. The hardness of water is due to..........metal ions

(A) Ca^{2+} and Na^+ (B) Mg^{2+} and K^+ (C) Ca^{2+} and Mg^{2+} (D) Zn^{2+} and Ba^{2+}

HY0022

23. The formula of heavy water is :-

(A) H_2O^{18} (B) D_2O (C) T_2O (D) H_2O^{17}

HY0023

24. Pure de-mineralised water can be obtained by -

(A) Na^+ cation exchanger and Cl^- anion exchanger

(B) H^+ cation exchanger only

(C) H^+ cation exchanger and OH^- anion exchanger

(D) Na^+ cation exchanger only

HY0024

HYDROGEN PEROXIDE (H_2O_2)

25. The bleaching properties of H_2O_2 are due to its :-

(A) Reducing properties (B) Oxidising properties

(C) Unstable nature (D) Acidic nature

HY0025

26. Hydrogen peroxide has a :-

(A) Linear structure (B) Pyramidal structure

(C) Closed book type structure (D) Half open book type structure

HY0026

27. Hydrogen peroxide is a :-

(A) Liquid (B) Gas (C) Solid (D) Semi-solid

HY0027

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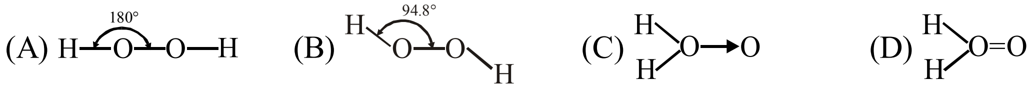
28. Which of the following is a true structure of H_2O_2



HY0028

29. Decomposition of H_2O_2 is retarded by :-

$2H_2O_2(\ell) \rightarrow 2H_2O(\ell) + O_2(g)$

(A) Acetanilide
(B) MnO_2
(C) Zinc
(D) Finely divided metals

HY0029

30. H_2O_2 is :-

(A) An oxiding agent
(B) Both oxidising and reducing agent
(C) Reducing agent
(D) None of the above

HY0030

31. H_2O_2 is always stored in black bottles because :-

(A) It is highly unstable
(B) Its enthalpy of decomposition is high
(C) It undergoes auto-oxidation on prolonged standing
(D) None of these

HY0031

32. Acidified solution of $K_2Cr_2O_7$ on treatment with H_2O_2 yields :-

(A) $CrO_3 + H_2O + O_2$
(B) $Cr_2O_3 + H_2O + O_2$
(C) $CrO_5 + H_2O + K_2SO_4$
(D) $H_2Cr_2O_7 + H_2O + O_2$

HY0032

33. H_2O_2 restores the colour of old lead paintings, blackened by the action of H_2S gas by :-

(A) Converting PbO_2 to Pb
(B) By oxidising PbS to $PbSO_4$
(C) Converting $PbCO_3$ to Pb
(D) Oxidising $PbSO_3$ to $PbSO_4$

HY0033

34. The reaction, $H_2S + H_2O_2 \longrightarrow S + 2H_2O$ manifests:

(A) Acidic nature of H_2O_2
(B) Alkaline nature of H_2O_2
(C) Oxidising nature of H_2O_2
(D) Reducing nature of H_2O_2

HY0034

35. Hydrogen peroxide is now generally prepared on industrial scale by the :-

(A) Action of H_2SO_4 on barium peroxide
(B) Action of H_2SO_4 on sodium peroxide
(C) auto-oxidation of 2-alklylanthraquinols
(D) Burning hydrogen in excess of oxygen

HY0035

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EXERCISE # O-2

1. Which of the following is / are same for Ortho and Para hydrogen :-

(A) In the number of protons

(B) In the molecular mass

(C) In the nature of spins of nucleus

(D) In the nature of spins of electrons

HY0036

2. In Bosch's process which gas is **NOT** utilised for the production of hydrogen :-

(A) Producer gas (B) Water gas

(C) Coal gas (D) Natural gas

HY0037

3. The gas(es) used in the hydrogenation of oils in presence of nickel as a catalyst is / are :-

(A) Methane (B) Ethane (C) Ozone (D) Hydrogen

HY0038

4. Water softening by Clarke's process does **NOT** uses :-

(A) Calcium bicarbonate (B) Sodium bicarbonate

(C) Potash alum (D) Calcium hydroxide

HY0039

5. Which of the following produces hydrolith with dihydrogen :-

(A) Mg (B) Al (C) Cu (D) Ca

HY0040

6. Which process is/are used to remove permanent hardness :-

(A) Boiling (B) Clark's method

(C) On reaction with with NaOH (D) Permutit process

HY0041

7. Ionic hydrides is/are usually :-

(A) Good electrically conductors when solid (B) Easily reduced

(C) Good reducing agents (D) Liquid at room temperature

HY0042

8. Which of the following will produce hydrogen gas :-

(A) Reaction between Fe and dil. HCl

(B) Reaction between Zn and conc. H_2SO_4

(C) Reaction between Zn and NaOH

(D) Electrolysis of NaCl (aq.) in Nelson's cell

HY0043

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9. Ortho-hydrogen and para-hydrogen resembles in which of the following property :-

(A) Thermal conductivity
(B) Magnetic properties
(C) Chemical properties
(D) Heat capacity

HY0044

10. Which of the following statements concerning protium, deuterium and tritium is / are true :-

(A) They are isotopes of each other
(B) They have similar electronic configurations
(C) They exist in the nature in the ratio of 1 : 2 : 3
(D) Their mass numbers are in the ratio of 1 : 2 : 3

HY0045

11. Ionic hydrides are formed by :-

(A) Transition metals
(B) Elements of very high electropositivity
(C) Elements of very low electropositivity
(D) Metalloids

HY0046

12. Which of the following statements is/are correct :

(A) Atomic hydrogen is obtained by passing hydrogen gas through an electric arc
(B) 30% (w/v) or 100V H_2O_2 solution is not called per hydrol.
(C) Finely divided palladium absorbs large volume of hydrogen gas.
(D) Ortho and para hydrogen have same physical properties.

HY0047

13. Which hydride is/are an ionic hydride :-

(A) NH_3 (B) H_2S (C) $TiH_{1.73}$ (D) NaH

HY0048

14. Which of the following hydride is/are "electron-precise' type ?

(A) HF (B) H_2O (C) SiH_4 (D) PH_3

HY0049

15. Hydrogen peroxide can act as a :-

(A) A reducing agent
(B) An oxidising agent
(C) A dehydrating agent
(D) A bleaching agent

HY0050

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EXERCISE # S-1

Integer Type

1. Find out the sum of protons, electrons and neutrons in the heaviest isotope of hydrogen.

HY0051

2. Find out the number of following orders which are correct against the mentioned properties :

(i) $H_2 < D_2 < T_2$ (Number of protons)

(ii) $H_2 < D_2$ (Bond energy)

(iii) $H_2 < D_2 < T_2$ (Boiling point)

(iv) $H_2 < D_2 < T_2$ (No. of neutrons)

HY0052

3. Find out the number of following orders which are **NOT** correct against the mentioned properties :

(i) $CaH_2 < BeH_2$ (Electrical conductance in molten condition)

(ii) $LiH < NaH < CsH$ (Ionic character)

(iii) $H_2 < D_2 < F_2$ (Bond dissociation enthalpy)

(iv) $NaH < MgH_2 < H_2O$ (Reducing property)

HY0053

4. What is the oxidation state of oxygen of H_2O_2 in the final product when it reacts with ClO_3^-.

HY0054

5. Find out the value of 'x' in ion $[H_xO_4]^+$:

HY0055

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EXERCISE # S-2

Matrix Match Type

1. Match List I with List II and select the correct answer using the codes given below the lists :-

List – I	**List – II**
P. Heavy water	1. Bicarbonates of Mg and Ca in water
Q. Temporary hard	2. No foreign ions in water
R. Soft water	3. D_2O
S. Permanent hard	4. Sulphates and chlorides of Mg and Ca in water

Code :

	P	**Q**	**R**	**S**
(A)	3	1	2	4
(B)	3	4	2	3
(C)	3	2	1	4
(D)	2	3	1	4

HY0056

2. Match List I with List II and select the correct answer using the codes given below the lists :-

List – I	**List – II**
P. Calgon	1. Metallic hydride
Q. Non-stoichiometric compound	2. Polymetaphosphate of sodium
R. Covalent hydride	3. Hydrolith
S. Salt-like hydride	4. Hydrogen compounds of non-metals

Code :

	P	**Q**	**R**	**S**
(A)	2	1	3	4
(B)	3	4	2	3
(C)	2	1	4	3
(D)	2	3	1	4

HY0057

Comprehenstion Type :

Passage for Q.3 to Q.5

Hydrogen accounts for approximately 75% of the mass of the universe. Hydrogen serves as the nuclear fuel of our Sun and other stars, and these are mainly composed of hydrogen.

Hydrogen has three isotopes : hydrogen or protium (${}^{1}_{1}H$), deuterium or heavy hydrogen (D or ${}^{2}_{1}H$), tritium (T or ${}^{3}_{1}H$).

3. Which of the following is radioactive in nature ?

(A) hydrogen only
(B) deuterium only
(C) tritium only
(D) deuterium and tritium

HY0058

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4. Hydrogen, H_2, is very less abundant in the atmosphere due to -

(A) inflammable nature of H_2

(B) weak earth's gravity which is not able to hold light H_2 molecules

(C) diatomic nature of hydrogen

(D) very rapid reaction between hydrogen and atmospheric oxygen

HY0058

5. Liquid H_2 has been used as rocket fuel as

(A) its reaction with oxygen is highly exothermic

(B) it occupies small space

(C) it has high thrust

(D) all of the above

HY0058

MATCHING LIST TYPE 1 × 3 Q. (THREE LIST TYPE Q.)

Column - I Name	Column - II Formula	Column - III Specification
(1) Calogen	(P) $Na_6P_6O_{18}$	(i) Used to Remove temporary Hardness
(2) Permutit	(Q) $Na_2Al_2Si_2O_8.xH_2O$	(ii) Used to remove permanent hardness
(3) Perhydrol	(R) '100 V' H_2O_2	(iii) Used in Rockect propelent
(4) Washing Soda	(S) $Na_2CO_3.10H_2O$	(iv) Also named as zeolite

6. Which combination is **NOT** related to removal of Ca^{+2}/Mg^{+2} form the sample of water

(A) (1)-(P)-(i)(ii) (B) (2)-(Q)-(i)(ii)(iv) (C) (3)-(R)-(i)(ii) (D) (4)-(S)-(i)(ii)

HY0059

7. Which of the following is **INCORRECT** between column I & II

(A) 1–P (B) 2–Q (C) 3–R (D) 2–S

HY0059

8. Which of the following is **INCORRECT** matching between column III & column II

(A) (iii) – R (B) (iv) – Q (C) (iv) – P (D) none of these

HY0059

EXERCISE # JEE-MAIN

1. Which one of the following processes will produce hard water :- **[AIEEE 2003]**

(1) Saturation of water with $CaSO_4$ (2) Addition of Na_2SO_4 to water

(3) Saturation of water with $CaCO_3$ (4) Saturation of water with $MgCO_3$

HY0060

2. Very pure hydrogen (99.9%) can be made by which of the following processes? **[AIEEE 2012]**

(1) Reaction of salt like hydrides with water

(2) Reaction of methane with steam

(3) Mixing natural hydrocarbons of high molecular weight

(4) Electrolysis of water

HY0061

3. In which of the following reaction H_2O_2 acts as a reducing agent ? **[JEE(Main) 2014]**

(a) $H_2O_2 + 2H^+ + 2e^- \rightarrow 2H_2O$ (b) $H_2O_2 - 2e^- \rightarrow O_2 + 2H^+$

(c) $H_2O_2 + 2e^- \rightarrow 2OH^-$ (d) $H_2O_2 + 2OH^- - 2e^- \rightarrow O_2 + 2H_2O$

(1) (a), (c) (2) (b), (d) (3) (a), (b) (4) (c), (d)

HY0062

4. Which of the following statements about Na_2O_2 is **not** correct ? **[JEE(Main) 2014]**

(1) Na_2O_2 oxidises Cr^{3+} to CrO_4^{2-} in acid medium

(2) It is diamagnetic in nature

(3) It is the super oxide of sodium

(4) It is a derivative of H_2O_2

HY0063

5. Hydrogen peroxide acts both as an oxidising and as a reducing agent depending upon the nature of the reacting species. In which of the following cases H_2O_2 acts as a reducing agent in acid medium ? :- **[JEE(Main)Online-2014]**

(1) MnO_4^- (2) SO_3^{2-} (3) KI (4) $Cr_2O_7^{2-}$

HY0064

6. Permanent hardness in water cannot be cured by: **[JEE(Main)Online-2015]**

(1) Treatment with washing soda (2) Calgon's method

(3) Boiling (4) Ion exchange method

HY0065

7. From the following statements regarding H_2O_2, choose the incorrect statement :

(1) It has to be stored in plastic or wax lined glass bottles in dark **[JEE(Main)Online-2015]**

(2) It has to be kept away from dust

(3) It can act only as an oxidizing agent

(4) It decomposes on exposure to light

HY0066

8. In which of the following reaction, hydrogen peroxide acts as an oxidizing agent ?

(1) $I_2 + H_2O_2 + 2OH^- \rightarrow 2I^- + 2H_2O + O_2$ **[JEE(Main)-2017]**

(2) $HOCl + H_2O_2 \rightarrow H_3O^+ + Cl^- + O_2$

(3) $PbS + 4H_2O_2 \rightarrow PbSO_4 + 4H_2O$

(4) $2MnO_4^- + 3H_2O_2 \rightarrow 2MnO_2 + 3O_2 + 2H_2O + 2OH^-$

HY0067

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9. Hydrogen peroxide oxidises $[Fe(CN)_6]^{4-}$ to $[Fe(CN)_6]^{3-}$ in acidic medium but reduces $[Fe(CN)_6]^{3-}$ to $[Fe(CN)_6]^{4-}$ in alkaline medium. The other products formed are, respectively :

(1) $(H_2O + O_2)$ and $(H_2O + OH^-)$ (2) H_2O and $(H_2O + O_2)$

(3) H_2O and $(H_2O + OH^-)$ (4) $(H_2O + O_2)$ and H_2O **[JEE(Main) -2018]**

HY0068

10. The chemical nature of hydrogen preoxide is :- **[JEE(Main) -2019]**

(1) Oxidising and reducing agent in acidic medium, but not in basic medium.

(2) Oxidising and reducing agent in both acidic and basic medium

(3) Reducing agent in basic medium, but not in acidic medium

(4) Oxidising agent in acidic medium, but not in basic medium.

HY0069

11. The total number of isotopes of hydrogen and number of radioactive isotopes among them, respectively, are : **[JEE(Main) -2019]**

(1) 2 and 0 (2) 3 and 2 (3) 3 and 1 (4) 2 and 1

HY0070

12. Th correct statements among (a) to (d) regarding H_2 as a fuel are : **[JEE(Main) -2019]**

(a) It produces less pollutant than petrol

(b) A cylinder of compressed dihydrogen weighs $\sim 30\,\text{times}$ more than a petrol tank producing the same amount of energy

(c) Dihydrogen is stored in tanks of metal alloys like $NaNi_5$

(d) On combustion, values of energy released per gram of liquid dihydrogen and LPG are 50 and 142 kJ, respectively

(1) b and d only (2) a, b and c only (3) b, c and d only (4) a and c only

HY0071

13. The correct statements among (a) to (b) are: **[JEE(Main) -2019]**

(a) saline hydrides produce H_2 gas when reacted with H_2O.

(b) reaction of $LiAH_4$ with BF_3 leads to B_2H_6.

(c) PH_3 and CH_4 are electron - rich and electron-precise hydrides, respectively.

(d) HF and CH_4 are called as molecular hydrides.

(1) (c) and (d) only (2) (a), (b) and (c) only

(3) (a), (b), (c) and (d) (4) (a), (c) and (d) only **HY0072**

14. In comparison to the zeolite process for the removal of permanent hardness, the synthetic resins method is : **[JEE(Main) -2020]**

(1) less efficient as it exchanges only anions

(2) more efficient as it can exchange only cations

(3) less efficient as the resins cannot be regenerated

(4) more efficient as it can exchange both cations as well as anions

HY0073

15. Hydrogen has three isotopes (A), (B) and (C). If the number of neutron(s) in (A), (B) and (C) respectively, are (x), (y) and (z), the sum of (x), (y) an (z) is : **[JEE(Main) -2020]**

(1) 4 (2) 3 (3) 2 (4) 1

HY0074

16. Among the statements (a) - (d), the correct ones are - **[JEE(Main) -2020]**

(a) Decomposition of hydrogen peroxide gives dioxygen

(b) Like hydrogen peroxide, compounds , such as $KClO_3$, $Pb(NO_3)_2$ and $NaNO_3$ when heated liberated dioxygen

(c) 2-Ethylanthraquinone is useful for the industrial preparation of hydrogen peroxide.

(d) Hydrogen peroxide is used for the manufacture of sodium perborate

(1) (a), (b) and (c) only (2) (a) and (c) only

(3) (a), (b) , (c) and (d) (4) (a), (c) and (d) only

HY0075

17. Which of the following forms of hydrogen emits low energy β-particles? **[J-Main 2021]**

(1) Deuterium $^{2}_{1}H$ (2) Tritium $^{3}_{1}H$ (3) Protium $^{1}_{1}H$ (4) Proton H^+ **HY0081**

18. Given below are two statements : **[J-Main 2021]**

Statement I : H_2O_2 can act as both oxidising and reducing agent in basic medium.

Statement II : In the hydrogen economy, the energy is transmitted in the form of dihydrogen.

In the light of the above statements, choose the correct answer from the options given below :

(1) Both statement I and statement II are false **HY0082**

(2) Both statement I and statement II are true

(3) Statement I is true but statement II is false

(4) Statement I is false but statement II is true

19. The functional groups that are responsible for the ion-exchange property of cation and anion exchange resins, respectively, are : **[J-Main 2021]**

(1) $-SO_3H$ and $-NH_2$ **HY0083**

(2) $-SO_3H$ and $-COOH$

(3) $-NH_2$ and $-COOH$

(4) $-NH_2$ and $-SO_3H$

20. The single largest industrial application of dihydrogen is : **[J-Main 2021]**

(1) Manufacture of metal hydrides (2) Rocket fuel in space research **HY0084**

(3) In the synthesis of ammonia (4) In the synthesis of nitric acid

21. Which one of the following statements is incorrect ? **[J-Main 2021]**

(1) Atomic hydrogen is produced when H_2 molecules at a high temperature are irradiated with UV radiation. **HY0085**

(2) At around 2000 K, the dissociation of dihydrogen into its atoms is nearly 8.1%.

(3) Bond dissociation enthalpy of H_2 is highest among diatomic gaseous molecules which contain a single bond.

(4) Dihydrogen is produced on reacting zinc with HCl as well as NaOH(aq).

22. Dihydrogen reacts with CuO to give **[J-Main 2022]**

(1) CuH_2 (2) Cu (3) Cu_2O (4) $Cu(OH)_2$ **HY0086**

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EXERCISE # JEE ADVANCED

1. When zeolite (hydrated sodium aluminium silicate) is treated with hard water, the sodium ions are exchanged with :- **[IIT 1990]**

(A) H^+ ions (B) Ca^{2+} ions (C) SO_4^{2-} ions (D) OH^- ions

HY0076

2. Which of the following statement is correct :-

(A) Hydrogen has same ionisation potential as sodium

(B) H has same electronegativity as halogens

(C) It will not be liberated at anode

(D) H has oxidation state + 1, zero and – 1

HY0077

3. Polyphosphates are used as water softening agent becuase they :- **[IIT 2002]**

(A) Form soluble complexes with anionic species

(B) Precipitate anionic species

(C) Form soluble complexes with cationic species

(D) Precipitate cationic species.

HY0078

4. Hydrogen peroxide in its reaction with KIO_4 and NH_2OH respectively, is acting as a **[JEE Adv. 2014]**

(A) reducing agent, oxidising agent (B) reducing agent, reducing agent

(C) oxidising agent, oxidising agent (D) oxidising agent, reducing agent

HY0079

5. Which of the following combination will produce H_2 gas? **[JEE Adv. 2017]**

(A) Zn metal and NaOH(aq.) (B) Au metal and NaCN(aq.) in the presence of air

(C) Cu metal and conc. HNO_3 (D) Fe metal and conc. HNO_3

HY0080

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ANSWERS KEY

EXERCISE : O-1

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	A	D	C	D	A	C	A	A	D	B	B	C	A	A	B
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	D	B	C	A	C	A	C	B	C	B	D	A	B	A	B
Que.	31	32	33	34	35										
Ans.	C	C	B	C	C										

EXERCISE : O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,B,D	A,C,D	D	A,B,C	D	D	C	A,C,D	C	A,B,D
Que.	11	12	13	14	15					
Ans.	B	A,C	D	C	A,B,D					

EXERCISE : S-1

Que.	1	2	3	4	5
Ans.	4	3	3	0	9

EXERCISE : S-2

Que.	1	2	3	4	5	6	7	8
Ans.	A	C	C	B	D	C	D	C

EXERCISE : JEE-MAINS

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	4	2	3	1	3	3	3	2	2
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	3	2	3	4	2	3	2	2	1	3
Que.	21	22								
Ans.	2	B								

EXERCISE : JEE-ADVANCED

Que.	1	2	3	4	5
Ans.	B	D	C	A	A

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